

KV Responsible AI



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**Ethics Guidelines for AI
13.11.2023**



Mini Task 1: Short review

226 students submitted mini task 1

The task was to describe potential i) compute-related impacts, ii) immediate application impacts and iii) system-level impacts of a self-selected AI application (based on the Impact Assessment Framework by Kaack et al.).

Top selected AI applications:

- 1— Tesla Autopilot / Tesla Self-Driving Systems
- 2— Alpha Fold (Deep Mind by Alphabet)
- 3— IBM Watson
- 4— Alpha Go
- 5— Recommender-Systems (Spotify, Youtube, etc.)

Group task “dilemma discussion”: Update

- Self-assignment to a group was due on 10.11.
- Groups with too few members have now been merged, nobody had to change topic
- You should now be able to see your group-specific chatroom on Moodle (below “Overview of personal course attendance”)

Next steps:

- Get in touch with each other in the group chat by 17.11.!
- Determine who goes into the PRO and CON subgroups (half of the members each)
- Arrange dates for i) a subgroup meeting and ii) the dilemma discussion in the whole group

Detailed instructions for the group task are in the slides from 30.10.2023.

AI Ethics Guidelines

In recent years, many supranational and national organizations and companies have published principles for ethical/responsible AI. A few examples:

[OECD Principles on Artificial Intelligence](#)

[UNESCO Recommendation on the Ethics of Artificial Intelligence](#)

[WEF Recommendations on Responsible Generative AI \(World Economic Forum\)](#)

[EU Ethics Guidelines for Trustworthy AI](#)

[DoD's Ethical Principles for Artificial Intelligence \(U.S. Department of Defense\)](#)

[Google AI Principles](#)

Global AI race: Can ethics matter?

USA: Most AI companies/start-ups, most private equity and venture capital funding, talent attractor

Europe: High-quality AI talents and research, struggles to translate research into business applications, loses talent (mostly to the US)

China: Rapidly catches up, leads in adoption speed and available data (number of inhabitants), less strict data protection guidelines ("Social scoring")



Global AI race: Can ethics matter?

Developing a European way:

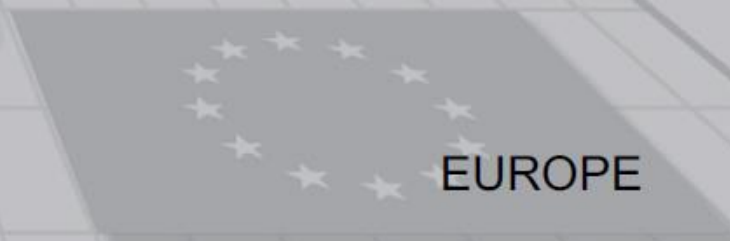
- The EU's coordinated plan on AI states that its “ambition is for Europe to become the world-leading region for developing and deploying **cutting-edge, ethical and secure AI.**”
- Positioning “AI made in Europe” as a positive global brand
- Ethics is an innovation enabler that offers competitive advantage: Trustworthy AI as economic value?



CHINA



USA



EUROPE

High-Level Expert Group on AI (European Commission)

The European Commission appointed a group of experts to provide advice on its AI strategy.

52 experts from industry, academia, and civil society who formed the High-Level Expert Group on AI (HLEG AI).

Deliverable 1:

Ethics Guidelines for Trustworthy AI

Deliverable 2:

Policy and Investment Recommendations for Trustworthy AI

Deliverable 3:

Assessment List for Trustworthy AI (ALTAI)

Deliverable 4:

Sectoral Considerations on the Policy and Investment Recommendations

EU Ethics Guidelines for Trustworthy AI

One of the core pieces of the European AI strategy are the Ethics Guidelines for Trustworthy AI that were published in April 2019.

The development of European AI should be

1. **Lawful**, respecting all applicable laws and regulations;
2. **Ethical**, respecting ethical principles and values (see 7 Key Factors); and
3. **Robust**, both technically and in terms of its social environment.

The guidelines were created by the 52 experts of the High-Level Expert Group on AI (HLEG AI).

The guidelines are meant to inform EU legislation (“AI Act”) and policies on AI.



7 Key Requirements for Trustworthy AI

7 key requirements for **ethical AI**:

Human agency and oversight

Technically robustness & safe

Privacy and data governance

Transparency

Diversity, non-discrimination and fairness

Societal and environmental wellbeing

Accountable

Will your algorithms pass the test?
Create AI humans can trust.

Reika Ala-Pietiläinen, 1981, provides a trustworthy AI
...
Privacy & data governance
Transparency
Diversity, non-discrimination & fairness
Societal & environmental well-being
Accountability
europa.eu/

#AI #ArtificialIntelligence

digitalDay2019

Andreas Antonopoulos, Member of the European Parliament and 3 others



1) Human agency and oversight

Key aspects:

1.1 Fundamental rights

Like many technologies, AI systems can equally enable and hamper fundamental rights (e.g. liberty, freedom of thought, freedom of expression, freedom of movement, self-determination, ...)

Mechanisms should be put into place to receive feedback regarding AI systems that potentially infringe on fundamental rights.

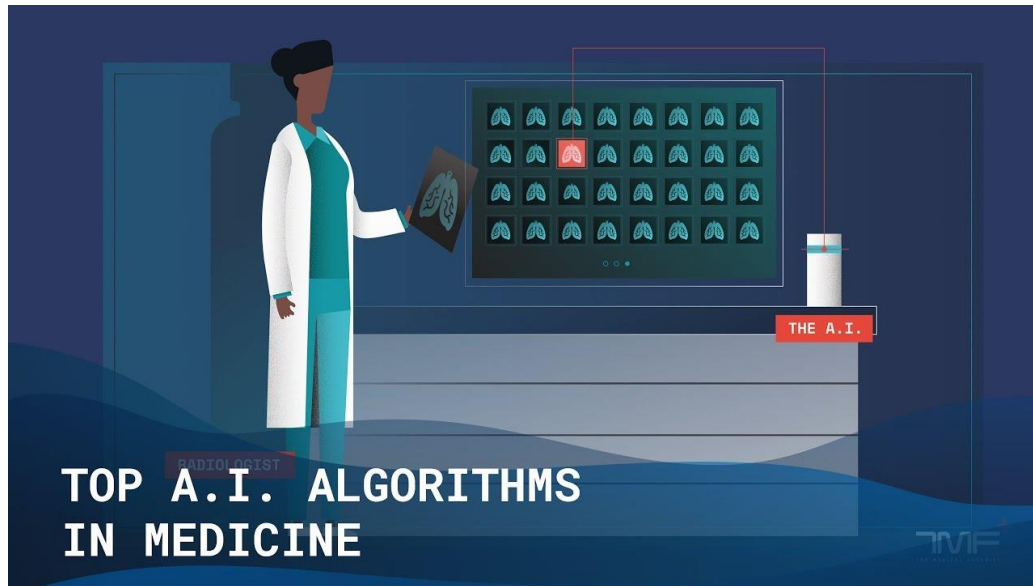
1.2 Human agency

Users should be able to make informed autonomous decisions regarding AI systems. They should be given the knowledge and tools to comprehend and interact with AI systems to a satisfactory degree and, where possible, be enabled to reasonably self-assess or challenge the system.

1.3 Human oversight

Oversight may be achieved through governance mechanisms such as a human-in-the-loop (HITL), human-on-the-loop (HOTL), or human-in-command (HIC) approach.

Algorithmic decision aids in relevant areas of life



2) Technical robustness & safety

Key aspects:

2.1 Resilience to attack and security

AI systems, like all software systems, should be protected against vulnerabilities that can allow them to be exploited by adversaries, e.g. hacking.

2.2 Fallback plan and general safety

AI systems should have safeguards that enable a fallback plan in case of problems. This can mean that AI systems switch from a statistical to rule-based procedure, or that they ask for a human operator before continuing their action.

2.3 Accuracy

Accuracy pertains to an AI system's ability to make correct judgements, for example to correctly classify information into the proper categories, or its ability to make correct predictions, recommendations, or decisions based on data or models.

3) Privacy and data governance

Key aspects:

3.1 Privacy and data protection

AI systems must guarantee privacy and data protection throughout a system's entire lifecycle. This includes the information initially provided by the user, as well as the information generated about the user over the course of their interaction with the system (e.g. outputs that the AI system generated for specific users or how users responded to particular recommendations).

3.2 Access to data

In any given organization that handles individuals' data (whether someone is a user of the system or not), data protocols governing data access should be put in place. These protocols should outline who can access data and under which circumstances.

3.3 Quality and integrity of data

The quality of the data sets used is paramount to the performance of AI systems. When data is gathered, it may contain socially constructed biases, inaccuracies, errors and mistakes. This needs to be addressed prior to training with any given data set.

Reproduction of Human Bias by ML (Google Translate, 2019)

[Anmelden](#)

Google Übersetzer

Sofortübersetzung deaktivieren

EnglischDeutschFranzösischTürkisch - erkannt



DeutschEnglischFranzösisch

Übersetzen

o bir asker
o bir öğretmen
o bir doktor
o bir hemşire

53/5000

he is a soldier
She's a teacher
he is a doctor
she is a nurse

4) Transparency

Key aspects:

4.1 Traceability

The data sets and the processes that yield the AI system's decision, including those of data gathering and data labelling as well as the algorithms used, should be documented to the best possible standard

4.2 Explainability

Technical explainability requires that the decisions made by an AI system can be understood and traced by human beings.

4.3 Communication

AI systems should not represent themselves as humans to users; humans have the right to be informed that they are interacting with an AI system.



5) Diversity, non-discrimination and fairness

Key aspects:

5.1 Avoidance of unfair bias

Identifiable and discriminatory bias should be removed in the collection phase where possible. Moreover, hiring from diverse backgrounds, cultures and disciplines can ensure diversity of opinions and should be encouraged.

5.2 Accessibility and universal design

Particularly in business-to-consumer domains, systems should be user-centric and designed in a way that allows all people to use AI products or services, regardless of their age, gender, abilities or characteristics.

5.3 Stakeholder Participation

In order to develop AI systems that are trustworthy, it is advisable to consult stakeholders who may directly or indirectly be affected by the system throughout its life cycle.

6) Societal and environmental wellbeing

Key aspects:

6.1 Sustainable and environmentally friendly AI

Research should be fostered into AI solutions addressing areas of global concern, such as the Sustainable Development Goals. AI systems promise to help tackling some of the most pressing societal concerns, yet it must be ensured that this occurs in the most environmentally friendly way possible.

6.2 Societal impact

While AI systems can be used to enhance social skills, they can equally contribute to their deterioration. This could also affect people's physical and mental wellbeing. The effects of these systems must therefore be carefully monitored and considered.

6.3 Society and Democracy

The use of AI systems should be given careful consideration particularly in situations relating to the democratic process, including not only political decision-making but also electoral contexts.

7) Accountability

Key aspects:

7.1 Auditability

Evaluation by internal and external auditors, and the availability of such evaluation reports, can contribute to the trustworthiness of AI. In applications potentially affecting fundamental rights, including safety-critical applications, AI systems should be able to be independently audited.

7.2 Minimisation and reporting of negative impacts

The use of impact assessments (e.g. red teaming or forms of Algorithmic Impact Assessment) both prior to and during the development, deployment and use of AI systems can be helpful to minimise negative impact. These assessments must be proportionate to the risk that the AI systems pose. Protection must be available for whistle-blowers, NGOs, trade unions or other entities when reporting legitimate concerns about an AI-based system.

7.3 Redress

When unjust adverse impact occurs, accessible mechanisms should be foreseen that ensure adequate redress. Knowing that redress is possible when things go wrong is key to ensure trust.

Critique on the EU's Ethics Guidelines ...

"It's really good news: Europe has just taken the lead in the hotly contested global debate on the ethics of artificial intelligence (AI). On Monday in Brussels, the EU Commission presented its Ethics Guidelines for Trustworthy AI. The 52-member High-Level Expert Group on Artificial Intelligence (HLEG AI), of which I am a member, worked on the text for nine months. The result is a compromise of which I am not proud, but which is nevertheless the best in the world on the subject. The United States and China have nothing comparable. How does it fit together?"

This phenomenon is an example of "ethics washing". Industry organizes and cultivates ethical debates to buy time – to distract the public and to prevent or at least delay effective regulation and policy-making. Politicians also like to set up ethics committees because it gives them a course of action when, given the complexity of the issues, they simply don't know what to do – and that's only human. At the same time, however, industry is building one "ethics washing machine" after another. Facebook has invested in the TU Munich – funding an institute to train AI ethicists. Similarly, until recently Google had engaged philosophers Joanna Bryson and Luciano Floridi for an "Ethics Panel", however this was abruptly discontinued at the end of last week.

TAGESSPIEGEL Suche Anmelden ABO

T+ Politik Internationales Berlin Gesellschaft Wirtschaft Kultur Wissen Gesundheit Sport Meinungen >

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Politik | EU guidelines: Ethics washing made in Europe

Anzeige



EU guidelines Ethics washing made in Europe

On Tuesday, the EU has published ethics guidelines for artificial intelligence. A member of the expert group that drew up the paper says: This is a case of ethical white-washing.

08.04.2019, 15:48 Uhr

KOMMENTARE

Thomas Metzinger is Professor of Theoretical Philosophy at the University of Mainz and was a member of the commission's expert group that has worked on the guidelines published on Tuesday.

Anzeige

TAGESSPIEGEL

Mini Task 2: Ethics Washing?

The case of Timnit Gebru and Google.

Many tech companies employ teams responsible for controlling and communicating “ethical” decisions and actions. While the work of such teams might be helpful for the company itself to control and communicate their social and environmental impact, their *actual* impact on company decisions may be limited. But what happens if someone who is directly responsible for Ethics of AI publicly criticizes their own company’s goals? Meet Timnit Gebru. The computer scientist co-lead a team on the Ethical AI at Google. In 2020, she was laid off by the company for trying to publish a scientific paper with a critical take on Large Language Models.

1—Watch the video on the case of Timnit Gebru and do your own research. How did News outlets and Social Media react to this case? What did Google say about it at the time?

2—Read the controversial paper “On the dangers of stochastic parrots: Can Language Models be too big?”

Hand in the two following files:

A— PDF Visualize the main controversy between Timnit Gebru and Google on a MindMap

B— VOICE RECORDING (max. 2 min.): Imagine you are responsible for handling this case at Google. How would you deal with the AI ethics department? Do you think Google was right to lay off Timnit Gebru? Shall the establishment of an Ethical AI Department at Google be seen as Ethics Washing? Describe at least two arguments to support your viewpoints.

Further information and resources on Moodle!

Deadline for submission: 27.11.2023, 23:59



Questions?

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See you next week!

